

Theory of Computation & Automata Theory

Extracted Video Titles | Total Videos: 114 | Export Date: Sep 5, 2026

#	Video Title	Link
1	Introduction to Theory of Computation	View
2	Finite State Machine (Prerequisites)	View
3	Finite State Machine (Finite Automata)	View
4	Deterministic Finite Automata (Example 1)	View
5	Deterministic Finite Automata (Example 2)	View
6	Deterministic Finite Automata (Example 3)	View
7	Deterministic Finite Automata (Example 4)	View
8	Regular Languages	View
9	Operations on Regular Languages	View
10	Non-Deterministic Finite Automata	View
11	Formal Definition of Non-Deterministic Finite Automata (NFA)	View
12	Non-Deterministic Finite Automata (Solved Example 1)	View
13	Non-Deterministic Finite Automata (Solved Example 2)	View
14	Non-Deterministic Finite Automata (Solved Example 3)	View
15	Conversion of NFA to DFA	View
16	Conversion of NFA to DFA (Example 1)	View
17	Conversion of NFA to DFA (Example 2)	View
18	Conversion of NFA to DFA (Example 3)	View
19	Conversion of NFA to DFA (Example 4)	View
20	Minimization of Deterministic Finite Automata (DFA)	View
21	Minimization of DFA (Example 1)	View
22	Minimization of DFA (Example 2)	View
23	Minimization of DFA (With Multiple Final States)	View
24	Minimization of DFA (With Unreachable States)	View
25	Myhill Nerode Theorem - Table Filling Method	View
26	Myhill Nerode Theorem - Table Filling Method (Example)	View
27	Finite Automata With Outputs	View
28	Construction of Mealy Machine	View
29	Construction of Mealy Machine (Example 1)	View
30	Construction of Mealy Machine (Example 2)	View
31	Construction of Moore Machine	View
32	Construction of Moore Machine (Example 1)	View

#	Video Title	Link
33	Construction of Moore Machine (Example 2)	View
34	Conversion of Moore Machine to Mealy Machine	View
35	Conversion of Moore Machine to Mealy Machine (Example 1)	View
36	Conversion of Moore Machine to Mealy Machine (Example 2)	View
37	Conversion of Mealy Machine to Moore Machine	View
38	Conversion of Mealy Machine to Moore Machine (Example 1)	View
39	Conversion of Mealy Machine to Moore Machine (Example 2)	View
40	Conversion of Mealy Machine to Moore Machine (Using Transition Table)	View
41	Epsilon NFA	View
42	Conversion of Epsilon NFA to NFA	View
43	Conversion of Epsilon NFA to NFA - Examples (Part 1)	View
44	Conversion of Epsilon NFA to NFA - Examples (Part 2)	View
45	Regular Expression	View
46	Regular Expression - Examples	View
47	Identities of Regular Expression	View
48	Arden's Theorem	View
49	An Example Proof using Identities of Regular Expressions	View
50	Designing Regular Expressions	View
51	NFA to Regular Expression Conversion	View
52	DFA to Regular Expression Conversion	View
53	DFA to Regular Expression Conversion (when the DFA has Multiple Final States)	View
54	Conversion of Regular Expression to Finite Automata	View
55	Conversion of Regular Expression to Finite Automata - Examples (Part 1)	View
56	Conversion of Regular Expression to Finite Automata - Examples (Part 2)	View
57	Conversion of Regular Expression to Finite Automata - Examples (Part 3)	View
58	Equivalence of Two Finite Automata	View
59	Equivalence of Two Finite Automata (Example)	View
60	Pumping Lemma (For Regular Languages)	View
61	Pumping Lemma (For Regular Languages) Example 1	View
62	Pumping Lemma (For Regular Languages) Example 2	View
63	Regular Grammar	View
64	Derivations from a Grammar	View

#	Video Title	Link
65	Context Free Grammar & Context Free Language	View
66	Regular Languages & Finite Automata (Solved Problem 1)	View
67	Regular Languages & Finite Automata (Solved Problem 2)	View
68	Regular Languages & Finite Automata (Solved Problem 3)	View
69	Regular Languages & Finite Automata (Solved Problem 4)	View
70	Regular Languages & Finite Automata (Solved Problem 5)	View
71	Regular Languages & Finite Automata (Solved Problem 6)	View
72	Method to find whether a String belongs to a Grammar or not	View
73	Derivation Tree (Left & Right Derivation Trees)	View
74	Ambiguous Grammar	View
75	Simplification of CFG (Reduction of CFG)	View
76	Simplification of CFG (Removal of Unit Productions)	View
77	Simplification of CFG (Removal of Null Productions)	View
78	Chomsky Normal Form & CFG to CNF Conversion	View
79	Conversion of CFG to Chomsky Normal Form	View
80	Greibach Normal Form & CFG to GNF Conversion	View
81	CFG to GNF Conversion (Removal of Left Recursion)	View
82	Pumping Lemma (For Context Free Languages)	View
83	Pumping Lemma (For Context Free Languages) - Examples (Part 1)	View
84	Pumping Lemma (For Context Free Languages) - Examples (Part 2)	View
85	Pushdown Automata (Introduction)	View
86	Pushdown Automata (Formal Definition)	View
87	Pushdown Automata (Graphical Notation)	View
88	Pushdown Automata Example - Even Palindrome (Part 1)	View
89	Pushdown Automata Example - Even Palindrome (Part 2)	View
90	Pushdown Automata Example - Even Palindrome (Part 3)	View
91	Equivalence of CFG and PDA (Part 1)	View
92	Equivalence of CFG and PDA (Part 2a)	View
93	Equivalence of CFG and PDA (Part 2b)	View
94	Turing Machine - Introduction (Part 1)	View
95	Turing Machine - Introduction (Part 2)	View
96	Turing Machine (Formal Definition)	View

#	Video Title	Link
97	Turing Machine (Example 1)	View
98	Turing Machine (Example 2)	View
99	The Church-Turing Thesis	View
100	Turing Machine for Even Palindromes	View
101	Turing Machine Programming Techniques (Part 1)	View
102	Turing Machine Programming Techniques (Part 2)	View
103	Turing Machine Programming Techniques (Part 3)	View
104	Multitape Turing Machine	View
105	Nondeterministic Turing Machine (Part 1)	View
106	Nondeterministic Turing Machine (Part 2)	View
107	Turing Machine as Problem Solvers	View
108	Decidability and Undecidability	View
109	Universal Turing Machine	View
110	The Halting Problem	View
111	Undecidability of the Halting Problem	View
112	The Post Correspondence Problem	View
113	Undecidability of the Post Correspondence Problem	View
114	TOC - Conclusion and Summary	View